Lesson 9-3 · Session 1

Equations from Tables

Name _____ Date _____ Period _____

Maryland Standard 6.AT.11 · Learning target — I can write an equation that matches a table of values.

Show a model, calculation, or explanation. Answer every part and label units when needed.

Exercises 1–12 earn 1 point for a correct response and 1 for supporting work. Spiral Review and Test Practice earn 1 point each.

Learn from a model

REMEMBER

- Look at a table of values to find the rule connecting x and y, then write it as an equation (y = …).
- Check your equation works for every pair in the table, not just one.
- Watch out — if the y-values do not start at zero when x = 0, the rule includes a constant.
- Check it — substitute two different rows into your equation before you trust it.

WORKED EXAMPLE

Writing an Equation from a Table A table shows x: 1, 2, 3, 4 hours worked and y: 18, 36, 54, 72 dollars earned. Write an equation.

Step 1 Define the variables: x = hours worked, y = total dollars earned.

Step 2 Check each row for one rate: 18 ÷ 1, 36 ÷ 2, and 54 ÷ 3 all give 18.

Step 3 The constant rate is 18 dollars per hour, so y is 18 times x.

Step 4 Write it: y = 18x. Test the last row: 18 × 4 72, which matches.

Answer y = 18x, where x is hours worked and y is total dollars earned.

Practice

1. Write an equation for the table: x: 1,2,3 y: 4,8,12. Equation: y = _____

Answer _____

SHOW YOUR THINKING

2. Write an equation for the table: x: 2,4,6 y: 5,7,9. Equation: y = _____

Answer _____

SHOW YOUR THINKING

3. Check that your equation from problem 2 works for all three pairs in the table.

Answer _____

SHOW YOUR THINKING

4. Write the equation that fits every row of each table.

a. x: 1, 2, 3 and y: 6, 12, 18 y = _____
b. x: 1, 2, 3 and y: 9, 10, 11 y = _____
c. x: 2, 3, 4 and y: 14, 21, 28 y = _____
d. x: 5, 6, 7 and y: 0, 1, 2 y = _____
e. x: 3, 6, 9 and y: 30, 60, 90 y = _____

Answer _____

SHOW YOUR THINKING

5. Bulk Rice. A market's table shows x: 4, 8, 12 pounds of rice and y: 12, 24, 36 dollars. Write the equation that relates x and y, then find the cost of 15 pounds.

Answer _____

SHOW YOUR THINKING

6. Band Fundraiser. A band's table shows x: 2, 5, 9 boxes of popcorn sold and y: 16, 40, 72 dollars raised. Write the equation, then find the money raised from 14 boxes.

Answer _____

SHOW YOUR THINKING

7. Two Tables. Table A shows x: 1, 2, 3 and y: 7, 14, 21. Table B shows x: 1, 2, 3 and y: 7, 8, 9. Write an equation for each table, then explain how the two rules are different. Sentence Frame: "I know this because..."

Answer _____

EXPLAIN WITH MATHEMATICAL EVIDENCE

8. Make your own table of values with three rows that follows a rule of the form y = (a number) times x. Write the equation, then show that it works for every row.

Answer _____

SHOW YOUR THINKING

9. For x: 1, 2, 3, 4 and y: 6, 12, 18, 24, write the equation.

Answer _____ the equation

SHOW YOUR THINKING

10. For x: 1, 2, 3 and y: 25, 50, 75, write the equation.

Answer _____ the equation

SHOW YOUR THINKING

11. With a Constant. For x: 1, 2, 3 and y: 11, 16, 21, write the equation.

Answer _____ the equation

SHOW YOUR THINKING

12. Find the Mistake. For x: 1, 2, 3 and y: 8, 16, 24, Rae wrote y = x + 7 because it works for the first row. Explain the mistake.

Answer _____ the mistake and the equation

EXPLAIN WITH MATHEMATICAL EVIDENCE

Spiral review

13. From Lesson 9-2 Session 2. A graph through (3, 15) shows what rate?

Answer _____

SHOW YOUR THINKING

14. From Lesson 8-5 Session 1. Does n > 2 use an open or a closed dot?

Answer _____

SHOW YOUR THINKING

15. From Lesson 6-7 Session 1. Find the GCF of 30 and 45.

Answer _____

SHOW YOUR THINKING

Test practice

16. For x: 1, 2, 3 and y: 9, 18, 27, the equation is —

□ A. y = 9x	□ B. y = x + 9
□ C. y = 9 + x	□ D. y = x ÷ 9

17. For x: 1, 2, 3 and y: 6, 9, 12, the equation is —

□ A. y = 6x	□ B. y = 3x + 3
□ C. y = x + 6	□ D. y = 3x

18. A constant in the rule shows up as a value when x equals —

□ A. the rate	□ B. 1
□ C. 0	□ D. the total

Check and revise

CHECK YOUR MATHEMATICS	HOW DID IT GO?
Check every given input-output pair, including any starting amount. Every part answered / Work shown / Units labeled when needed	I can do this: Independently / With a hint / With help Question to revisit: _____

One correction or question I want to discuss: _____

Substitute one input-output pair into your rule. Check units and whether your answer fits the situation.